



HomeoCure: LLM-RAG Framework for Symptom-Based Homeopathy Remedy Recommendation

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Abstract. Accurate homeopathic remedy recommendation is crucial for timely identification of disease and treatment planning. However, the prevailing approaches in the domain of homeopathic medicines recommendation are challenged due to presence of incomplete symptom information and lack of semantic understanding which lead to imprecise remedy suggestions. To overcome these limitations, the present work proposes Homeocure, a homeopathy remedy recommendation system based on the symptoms provided by means of integrating Large Language Models (LLMs) with a Retrieval-Augmented Generation (RAG) pipeline. To accomplish this task, firstly, the Homeopathic reference text is digitized using PyMuPDF and structured into JSON. Secondly, to mitigate the incomplete symptom description, the model utilizes an interactive re-questioning module. This process provides a well-defined symptom for effective recommendation. Finally, sentenceTransformers have been incorporated for symptom embeddings and indexed using FAISS for rapid semantic similarity search. To conclude, retrieved symptom–remedy pairs are provided as context to a lightweight generative model (TinyLlama-1.1B-Chat-v1.0). To enhance the reliability, three metrics: embedding-based cosine similarity, TF-IDF similarity, and fuzzy string matching are applied to evaluate consistency between user queries and recommended remedies. Experimental results demonstrate that the system delivers interpretable, context-grounded, and computationally efficient recommendations, highlighting the potential of LLM + RAG architectures for decision support in homeopathy and related medical domains.

Keywords: Homeopathy · Symptom-based Recommendation · Retrieval-Augmented Generation (RAG) · Large Language Models (LLMs) · Semantic Search · Sentence Transformers · FAISS (Facebook AI Similarity Search) · Natural Language Processing (NLP) · Fuzzy Matching · Medical Decision Support

1 Introduction

Homeopathy is an alternative medicine in which remedies are prepared from natural extracts. The medicines are prescribed only after closely analyzing the symptoms reported by the patient. For analyzing the symptoms minutely, domain experts are

required. These experts are required to review hundreds of remedies to match with the defined symptoms before arriving at a decision which make the process very time consuming. Moreover, the experts are not available all the time at every place. In suburban areas and rural areas, the approachability to these practitioners is limited due to availability and affordability. These challenges can be addressed using an automated and intelligent systems that can provide symptom-based remedy recommendations in a reliable and interpretable manner.

Some researchers are working towards automated system for alternative medicines. These systems mainly rely on rule-based retrieval, keyword-based search, or classification methods. Though, these methods can prescribe the remedies to the patients but they often fail to capture semantic meaning and context hidden in the symptoms specified by the patient. For example, a patient describing “pulsating pain in the temples” may be looking for remedies related to headaches, but a keyword-based system might not be able to find exact keywords in the reference text and may result in wrong suggestion. The lack of semantic understanding can reduce the accuracy of recommendations and limits their practical use. Further, even a small symptom can change the course of treatment in homeopathy. These small symptoms may be missed by patients but are mentioned in the books. So, interactive re-questioning may be required which is not addressed in most of the existing systems.

Evolution of Large Language Models (LLMs) provides an opportunity to address these limitations. The advantages of LLMs are understanding context, semantics, and variation in natural language. Also, while Retrieval-Augmented Generation (RAG) frameworks ensure that outputs are generated using domain-specific knowledge. By combining these two approaches, it becomes possible to build intelligent assistants to retrieve relevant and personalized remedies. Using these technologies into homeopathy lets us automate classical homeopathy text and create systems that work for both practitioners and patients.

This paper introduces HomeoCure, that is a homeopathy recommendation tool which used large language model and a retrieval augmented generation (RAG) pipeline. The end goal is to give accurate and reliable remedy suggestion. The working includes the digitization of traditional homeopathy texts with *PyMuPDF* and organize everything into json file. Each entry lists remedies and their symptoms in clear categories. Then, we use *SentenceTransformers* (all-MiniLM-L6-v2) to turn those symptom descriptions into embeddings, which we index with FAISS for easy searching. When someone describes their symptoms, the system finds the top matches—symptom-remedy pairs that fit best. It uses these as context for TinyLlama-1.1B-Chat-v1.0, our lightweight generative model. If the model can’t make a clear recommendation, it asks the patient follow-up questions to fill in the gaps, then suggests the right remedy. The system ensures that the answer given is taken from trusted sources.

To enhance reliability, a validation layer has been added that checks results using cosine similarity, TF-IDF similarity, and fuzzy string matching. These metrics work together to ensure the recommendation actually matches what the user needs, boosting both accuracy and transparency. The entire pipeline is designed to be lightweight so that results can be delivered on time even on limited hardware. A user-friendly interface has

been made where user can input the symptoms and HomeCure quickly gives remedy suggestion that have already been verified and validated.

The key contributions of this work are summarized as follows:

1. Digitization and Structuring of homeopathy text: Conversion of unstructured homeopathic texts into a machine-readable JSON format for downstream NLP processing.
2. Semantic Symptom Retrieval: Use of sentence embeddings and FAISS indexing to enable fast and context-aware retrieval of symptom–remedy pairs.
3. Evidence-Grounded Recommendation Generation: Deployment of *TinyLlama* as a constrained generative model that provides concise, context-based recommendation.
4. Interactive re-questioning: An interactive re-questioning module is incorporated to handle incomplete or vague symptom descriptions.
5. Multi-Metric Validation Framework: Integration of cosine similarity, TF-IDF, and fuzzy matching to cross-check and validate retrieved remedies, improving confidence in results.
6. Operational Prototype: Development of a lightweight, explainable, and scalable system that demonstrates the practical feasibility of applying LLM + RAG architectures in the domain of homeopathy.

The remainder of the paper is organized as follows. Section 2 reviews existing work medical recommendation systems, including applications in homeopathy and related healthcare domains. Section 3 presents the methodology, including data preprocessing, semantic embedding, retrieval, generative modeling, and validation; followed by results and discussion in Sect. 4. Section 5 concludes the paper with key findings and outlines directions for future work.

2 Related Work

The integration of Artificial Intelligence (AI) and Natural Language Processing (NLP) into healthcare has seen rapid progress, with research focusing on decision support systems, retrieval-augmented generation (RAG) pipelines, and domain-specific medical chatbots. AI has extensive use in healthcare domain [12, 16]. While much of this work targets allopathic medicine, several attempts are made to extend AI-based methods to alternative medicine domains, including homeopathy.

2.1 AI-Driven Medical Decision Support

Hiesinger et al. (2023) propose Almanac, a RAG-based framework for clinical medicine, which ensures factual grounding of generative models using domain-specific retrieval [2]. By conditioning generative models on retrieved evidence, Almanac demonstrated the potential to improve the accuracy in medical reasoning tasks. Bhatt and Vaghela (2024) develops a Med-Bot, an AI assistant that gives users precise medical information by using a structured symptom analysis [1]. Their research shows what conversational agents can do, but the model depends heavily on structured data and don't use semantic retrieval.

Sree et al. (2024) take a different route. Their diagnostic chatbot leans on semantic similarity, using cosine similarity and fuzzy logic to validate its results. This makes its diagnoses more reliable [3]. Neupane et al. (2024) put forward MedInsight, a system that boosts patient queries with context from multiple sources, leading to better responses [4]. These all studies stressed how important it is to mix retrieval with validation. Still, there aim was for general medical use, not for homeopathy specifically.

2.2 AI in Alternative Medicine and Homeopathy

AI in homeopathy is still in its early days. Agarwal and Jegadeesan et.al [5, 6]. worked on homeopathy chatbots that try to offer personalized advice, but their systems depend mostly on rule-based logic or keyword matching. That really limits how well they understand context or adapt to new situations. Patil et al. (2024) took things a step further with Ayurveda, building AyUR-Bot to digitize classical texts and use AI for recommendations [11]. It shows you can bring ancient medical wisdom into the world of NLP, though the project still misses out on more complex validation and deep semantic understanding.

Yang et.al CaseGPT brings in a retrieval-based approach for medical case reasoning [7]. Jin et al. (2024) Health-LLM stress for even more personalized healthcare responses by building individual embeddings for users [8]. Personalization keeps people engaged, but it also demands a lot more from the computers running these systems. Su et.al (2024) showed that mixing extractive and abstractive summarization helps handle medical texts better [9]. Chin et al. proposed OYEN chatbot which stands out for using multilingual, domain-specific generative AI, which makes healthcare tools more accessible and inclusive [10].

A comparative analysis of recent related works is presented in Table 1, highlighting their objectives, strengths, and limitations.

Table 1. Comparative study of Related Work

Study / System	Domain	Approach	Strengths	Limitations
Med-Bot (Bhatt & Vaghela, 2024)	General medicine	Conversational AI + structured symptom analysis	Reliable responses; structured pipeline	Limited semantic understanding; no validation
Almanac (Hiesinger et al., 2023)	Clinical medicine	RAG with LLMs	Factual grounding; domain-specific retrieval	High computational requirements
Diagnostic Chatbot (Sree et al., 2024)	General medicine	Semantic matching + multi-metric validation	Improved contextual accuracy; validation layer	Not domain-specific; limited interpretability
MedInsight (Neupane et al., 2024)	General healthcare	Multi-source retrieval + LLM	Rich contextual responses	Broad scope; less interpretable for niche domains

(continued)

Table 1. (continued)

Study / System	Domain	Approach	Strengths	Limitations
Homeopathy Chatbots (Agarwal et al., Jegadeesan et al.)	Homeopathy	Rule-based / keyword retrieval	Personalized guidance	No semantic embeddings; low scalability
AyUR-Bot (Patil et al., 2024)	Ayurveda	Generative AI + digitized texts	Adaptation of ancient knowledge; explainability	No semantic validation; single-domain focus
CaseGPT (Yang, 2024)	Medical reasoning	Case-based RAG	Evidence-grounded outputs; reduced hallucinations	Computationally heavier; no symptom clustering
Health-LLM (Jin et al., 2024)	Healthcare prediction	Personalized RAG	Dynamic personalization	High complexity; scalability challenges

2.3 AI Based Homeopathy Tools

Now a days, many AI-powered tools for homeopathy have emerged. OpenAI has released a tool named Homeopathic Doctor which is based on ChatGPT [13]. Oorep provides a digital platform offering homeopathic resources and recommendations [15]. Nambisons proposed a tool called HomeoXpert [14], which demonstrate the growing interest in applying large language models (LLMs) to alternative medicine. As these tools highlight the potential of LLMs for homeopathy, they basically perform as conversational agents without clear mechanisms for semantic retrieval and interactive clarification. Given a symptom, these tools list down a number of medicines which may lead to confusion for the user. In contrast, the proposed system grounds its outputs in homeopathy reference text, validates recommendations with multiple similarity measures, and engages users in refining incomplete symptom queries.

Discussion: One of the possible limitations of these approaches is that they consider exact keyword-based matching and lack the semantic similarity criteria which results in limited accuracy.

3 HomeoCure: Proposed Methodology

In this section, we present the proposed framework, HomeoCure, which is homeopathy remedy suggestion system using LLM and RAG. Various steps involved in HomeoCure are data extraction, text structuring, semantic retrieval of the data, query processing, interactive re-questioning for query refinement and generative modeling. These steps are followed by multi-metric validation and user-friendly interface. Figure 1 illustrates the overall workflow. The details of each phase are presented below:

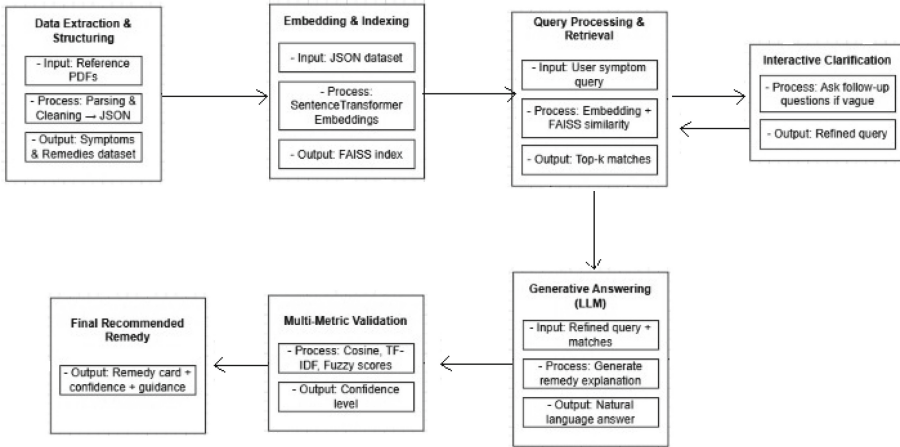


Fig. 1. Overall workflow of HomeoCure

3.1 Data Extraction and Structuring

Traditional reference text in homeopathy is written in unstructured narrative form, where remedies are described along with symptom categories such as *Mind*, *Head*, *Stomach*, *Extremities*, and so forth. While these texts are understandable, their narrative organization makes them difficult to query efficiently. Hence, to convert this text into a machine-readable format, we applied *PyMuPDF* (fitz) for PDF parsing. First, the preprocessing module filters out the irrelevant text from the PDF such as page headers, copyright statements or metadata. The text formatting errors such as line breaks and unnecessary symbols are also resolved to improve the quality of the extracted data. Next, the remedies and symptoms are identified using the heuristic rules. Medicine names are identified using regular expressions and remaining lines were treated as symptom entries. Also, null entries, malformed blocks, and duplicated content were removed after extraction. For each remedy, associated symptoms are segmented by category headings and grouped accordingly.

The preprocessed data is then organized in the JSON format as shown below:

Medicine name: Homeopathic remedy

Symptom categories: A list of symptoms categorized under various headings such as physical symptoms, emotional symptoms, etc.

This structured representation of the text allows to index the data in efficient manner so that the query can be processed faster and response can be generated effectively. It further allows the incremental expansion of the existing database, as new remedies and symptoms can be easily appended without modifying the existing schema.

3.2 Semantic Embedding and Indexing

The next step is to convert symptoms into dense vector embeddings that can capture the semantic meaning. For this, we have used *SentenceTransformers* (all-MiniLM-L6-v2).

It is a lightweight, high-performing transformer model which is trained for the semantic similarity tasks. Each symptom line is mapped into a 384-dimensional embedding vector. Next, these embeddings are indexed using FAISS (Facebook AI Similarity Search). It is an open-source library optimized for the nearest neighbor search in the high-dimensional space. Using FAISS, semantically similar entries across thousands of symptom lines can be searched in fraction of seconds. Unlike keyword matching, this search technique can allow us to map the query with its conceptually equivalent match. For example:

Query: “stomach pain with sour vomiting”

Symptom entry: “gastric distress then acid regurgitation”

It can be observed that these two descriptions have a very few common lexical tokens, even then they are matched as their embeddings lie close in vector space. This semantic approach provides significant improvement in recall and precision compared to rule-based methods.

3.3 Query Processing and Retrieval

When the user submits a query, first, the query is converted into a vector space using the same embedding model. Next, a k-nearest neighbor search is performed using the FAISS index to retrieve the top-k pairs <symptom, remedy> ranked according to their similarity. Each retrieved entry pair includes the remedy name with the exact symptom text from the reference text. Then, these retrieved entries are used for remedy recommendation for the given set of symptoms.

3.4 Interactive Re-questioning for Incomplete Symptoms

Interactive re-questioning module is one of the important phases of the proposed system. Generally, the user gives vague or incomplete symptoms, for instance, “*I have fever*” or “*I am feeling pain.*”, etc. Using these types of queries, it becomes difficult to match and identify remedies accurately, since homeopathy considered lots of information and detailing such as timing, intensity, duration, etc. for any recommendation. To address this gap, we first detect if the query is described in general and need further clarification. If yes, then, further questions are asked to the user. For example,

- User input: “I have fever.”
- System prompt: “Does the fever persist throughout the day, or does it appear mainly at night?”
- User refinement: “Mostly in the evening.”
- System recommends the remedies for evening-dominant fever.

This feature compliments the diagnostic questioning process actually used by homeopathic practitioners, where further detailing such as timing, duration, and aggravating or relieving factors are considered important for remedy selection. Incorporating re-questioning about symptoms can improve the precision and relevance of the prescribed remedies. It will make the system more closer to the clinical reasoning.

3.5 Generative Answering with LLM

Next, the relevant entries are retrieved using RAG based on the refined query. This data is further passed to the TinyLlama-1.1B-Chat-v1.0 model for the response generation. Unlike larger LLMs, *TinyLlama* works with a trade-off between efficiency and fluency, which makes it suitable for deployment in the constrained environment. The LLM always generate the responses strictly from the retrieved context. This mechanism increases the reliability of the response as it is reflected from the reference text only.

3.6 Validation

The proposed framework is validated using three metrics: Cosine similarity, TF-IDF and fuzzy matching. Cosine Similarity captures the semantic relationship between the query vector and the retrieved vectors. TF-IDF takes care of the statistical overlap in lexemes used in query and retrieved content, hence, it talks about the lexical similarity. Fuzzy Matching measures the closeness between the query and the symptom text even when there are some spelling or phrasing variations. It is used for approximate matching. Once these three scores are obtained, they are aggregated together using averaging to obtain the overall reliability score and to assign the rank to the remedies. The remedy with highest rank is recommended to the user.

4 Results and Discussion

The performance of the proposed system is measured in multiple stages. First, exploratory data analysis (EDA) of the digitized text is performed to understand the nature of the data. Then, a thorough evaluation is performed using three evaluation metrics.

4.1 Exploratory Data Analysis (EDA)

Once the given data is structured in the form of JSON format, it is analyzed to understand the distribution of remedies and symptoms.

- a. A word cloud is generated from all extracted symptom lines. As shown in Fig. 2, the most frequent terms are pain, cold, stool, etc. It can be observed that the preprocessed data contains both general and specialized symptoms. The visualization highlights the dominance of neurological, gastrointestinal, and reproductive symptoms, providing an overview of the data.



Fig. 2. Word cloud of symptoms

b. A bar graph (shown in Fig. 3) presents the top ten remedies with the largest number of associated symptoms. The remedies represented at X-axis demonstrates their broad coverage in the homeopathy reference material. It can be observed that that some remedies are more versatile and frequently prescribed. However, other medicines remain highly specific.

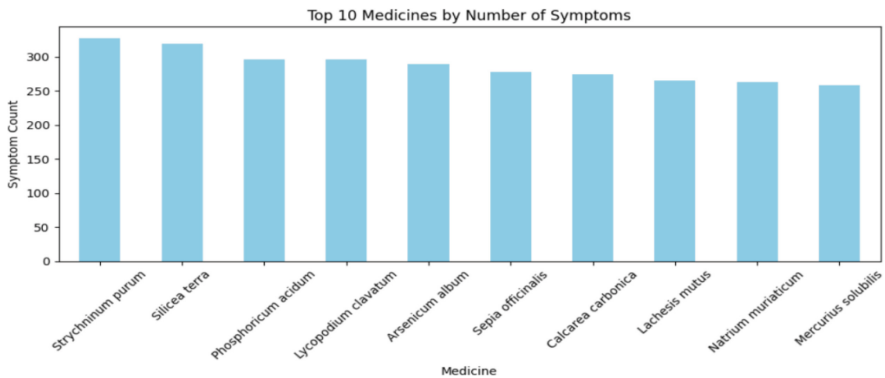


Fig. 3. Medicines by Symptom Count

- c. Figure 4 shows another bar graph which presents the frequency of symptom irrespective of remedies in the available dataset. This analysis validates that the dataset is not only comprehensive but also representative of real-world homeopathic practice.

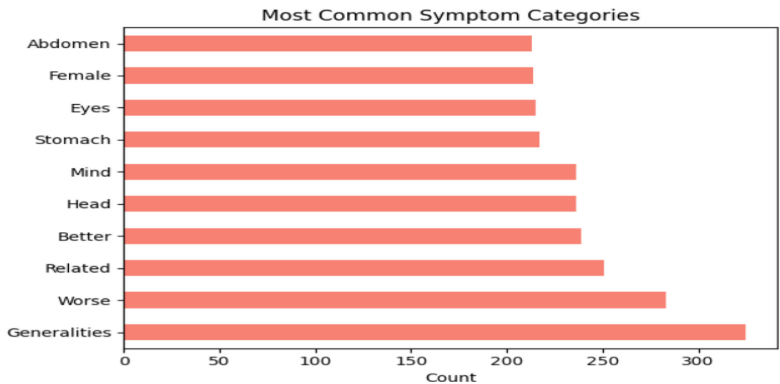


Fig. 4. Most Common Symptom Categories

It can be seen that the results presented by EDA provide a thorough understanding of the dataset. It ensures that the response retrieval and recommendation processes are based on well-structured, balanced dataset.

4.2 System Simulation and Evaluation

To validate performance of HomeoCure, following three metrics are used

- 1. Cosine Similarity (SentenceTransformers): It is used for confirming strong semantic alignment.
- 2. TF-IDF Cosine Similarity: It captures overlap in important terms, retrieves the semantic closer responses.
- 3. Fuzzy Matching (FuzzyWuzzy Partial Ratio): It applies approximate matching despite of variations in lexemes and phrasing.

Once, the values are obtained, simple averaging of the three values is used to find the rank. The remedy with the highest mean similarity score is assigned rank one. Table 2 presents a comparative evaluation and rank-wise remedies recommendation based on the aggregated scores obtained for the query “headache, nausea and dizziness” after interactive re-questioning. It can be seen that Iris versicolor is ranked one and recommended as the medicine to the user for the given set of symptoms.

Table 2. Comparative Evaluation of Different Validation metrics for “headache, nausea and dizziness”

Rank	Medicine	TF-IDF cosine	FuzzyWuzzy	Cosine Similarity (SentenceTransformers)	Rank Assigned
0	Iris versicolor	0.0454	0.75	0.2460	1
1	Kalium bichromicum	0.0196	0.75	0.2225	2

(continued)

Table 2. (continued)

Rank	Medicine	TF-IDF cosine	FuzzyWuzzy	Cosine Similarity (SentenceTransformers)	Rank Assigned
2	Ferrum metallicum	0.0218	0.75	0.2005	3
3	Borax veneta	0.0226	0.75	0.1209	4
4	Lappa arctium	0.0230	0.39	0.1726	5

5 Conclusion

This work presents a novel homeopathy symptom-based recommendation system that integrates Large Language Models (LLMs) with a Retrieval-Augmented Generation (RAG) framework, HomeoCure. First, the classical homeopathy reference text is digitized using PyMuPDF and then, structured into JSON format for efficient query processing. To handle incomplete or vague symptom description by user, the system incorporated a feature of an interactive re-questioning module that asks user with further questions for clarification of the symptoms. It ensures that the recommendations are based on well-defined detailed symptoms. Symptom embeddings are generated using SentenceTransformers (all-MiniLM-L6-v2) and indexed using FAISS for rapid semantic similarity search. Retrieved <symptom, remedy> pairs are used as context to generate the response using a lightweight generative model (TinyLlama-1.1B-Chat-v1.0).

The proposed system is evaluated using EDA, followed by three metric evaluation. The three metrics used are embedding-based cosine similarity, TF-IDF similarity, and fuzzy string matching. Finally, the scores of three metrics is aggregated. The remedy with high aggregated scores is recommended to the user.

In future, the work can be extended using more dynamic and flexible approach for re-questioning. Also, the recommendations are validated computationally. Still, clinically validation need to be done in real-world practitioner setting.

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